

problem 1: Missing Values

what did i find ?

I found many missing values in every column as the combined_task_1.csv didn't find some information

Where was it ?

The problem appeared in columns of task1_combined_data.csv
return_date, join_date, checkout_id, grade and publication_year.

How big was it ?

There are 91 missing values in the return date column.
There are 11 missing values in the join date column.
There are 26 missing values in the checkout ID column.
There are 41 missing values in the grade and 40 missing value publication year columns.

What did I do and why ?

in the return date column I replaced missing values with "Not Returned".

The reason for this is that the user might not have returned it yet.

in the date column I replaced values with "Unknown".

The reason for this is that it help me save the rest of information in this record and don't ignore the analyse.

in the checkout ID column I generated new ID starting after the highest ID in the column and signing it to all unkown id with diffrent id numbers.

The reason for this is to save the rest of information in this record with a unique id for every unkown id

For the grade and publication year columnsI left these values as normal.

The reason for this is that they may have some members not receive a grade if they have already graduated and oldbooks may not have a publication year in the and to Add values as mode or mean or medium may cause an error in analyse and give abnormal analyse and information.

problem 2: Invalid Member IDs

what did i find ?

I found borrowingrecords containing values that did not located in the original database .

Where was it ?

The problem was located in the first_name and last_name column.

How big was it ?

There were 5 records affected.

what did i do and why?

I deleted 5 rows completely from the dataset.

The reason for this is that these records contain users that didn't belong to dataset which may damage us in analyses.

problem 3: Duplicates

what did i find ?

I found records that were completely tepical .

Where was it ?

The duplication occurred at the record of the dataset .

How big it was

There were 8 duplicate records in the dataset.

What did I do and why ?

I used the `drop_duplicates()` function to remove them from the dataset.

The reason for this is that this records was typically equal so it may be duplicate by mistake and to help us in analyses

Issue 4: Inconsistent Text Formatting

what did i find ?

I found some text values written in formats, such as extra spacedifferences or in the design in general for the same word in thedataset.

Where was it ?

The issue appeared in the neighborhood and membership_status columns of thedataset.

How big was it ?

It was across the record of the neighborhood and membership_status columns which eyql 404 afer cleaning duplicates and missing values in the dataset.

What did I do and why ?

I used code to remove spaces and making the first letter of each word capital in the dataset.

The reason for this is to make the code read every neighborhood and membership_status correctly without thick that there is more than to types such as in membership_status.